Sitting on the Fence: Is Hedging Mediated by a Single Direction in Language Models?

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Abstract

Large language models (LLMs) are now widely used for advice and information, and many of the questions people ask them have no single correct answer. On such questions models often hedge: they say that "it depends" or that "there are valid arguments on both sides" instead of taking a position. Researchers have shown that another behaviour of chat models, refusing harmful requests, is mediated by a single direction in the residual stream. We ask whether hedging is mediated in the same way. For each model we label 373 evaluative questions by whether the model hedged or answered directly, take the difference between the mean activations of the two groups, and test the resulting direction on held-out questions against random directions and directions built from shuffled labels. In seven instruction-tuned LLMs from three families, removing the direction turns most hedged answers into direct ones, far more often than either control does. Adding it also makes models hedge, but less reliably. A hand-labelled sample confirms the effect of removal and suggests that the effect of addition is smaller than our automatic labels show. Within the Qwen2.5 family the direction is easier to detect in larger models and sits at a similar relative depth at every size. Removing it barely changes how often models refuse harmful requests, so hedging and refusal appear to rely on different directions.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Neural networks.*

Keywords

large language models, interpretability, activation steering, linear representations, hedging, refusal

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1 Introduction

Large language models (LLMs) are increasingly used as general assistants, and people bring them all kinds of questions, including many without a single correct answer: whether the death penalty should be abolished, or whether a university degree is worth what it costs. On questions like these, chat models often decline to take a side. They answer that the matter "depends on many factors" or that "there are valid arguments on both sides". We call this behaviour *hedging*, or fence-sitting. Hedging is not always a problem. Expressing uncertainty is often the right response when a model cannot be sure of an answer [19, 35]. But models also hedge on questions that do have a clear answer, and the habit matters for anyone trying to measure which positions these models hold, because surveys of the opinions and values of LLMs only work if the model commits to an answer in the first place [25, 26].

So far, this behaviour has mostly been studied by looking at what models say. Santurkar et al. [26] compare model answers on opinion surveys with those of human populations, and Röttger et al. [25] show that such results change when the model may answer freely instead of choosing from fixed options. Javed et al. [13] count how often models hedge on questions about human rights and find that the rate depends on which group the question names. Work on linguistic calibration asks whether the uncertainty a model expresses matches how often it is correct [19, 35]. These studies tell us when models hedge, but they treat the model as a black box. They say nothing about where in the network the decision to hedge is made.

Interpretability research offers a way to look inside. Several studies find that high-level concepts are represented as directions in the residual stream of transformer language models [17, 21]. A simple way to find such a direction is to run the model on two contrasting sets of inputs and take the difference between their mean activations. The direction can then be tested causally: adding

it to the activations, or removing it, should change the behaviour it is thought to represent [24, 31, 36]. Refusal is the best-known case. Arditi et al. [2] compared activations on harmful and harmless instructions and found, in 13 open chat models, a single direction whose removal makes the model comply with harmful requests and whose addition makes it refuse harmless ones. Later work showed that refusal can involve more than one dimension [33] and that models represent whether a request is harmful separately from the decision to refuse it [34].

Hedging resembles a mild form of refusal. The model does engage with the question, but it stops short of committing to an answer. This raises two questions that, as far as we know, no earlier work has addressed: is hedging also mediated by a single direction, and if so, is it the same direction as refusal? The answers matter beyond interpretability. A single hedging direction would give a simple handle for studying how readily a model takes positions, and knowing whether it overlaps with refusal tells us whether changing one behaviour would also change the other.

In this paper we adapt the refusal-direction method to hedging. Unlike refusal, hedging is not tied to a particular type of prompt, so we let each model define its own two groups: we ask it 373 evaluative questions and label each answer as hedged or direct. We build a direction from these labels, choose a layer and a strength on validation questions, and run the final test once on held-out questions. Since a difference of means produces a vector for any grouping of prompts, every test compares the hedging direction with random directions of the same size and with directions built from shuffled labels. We study seven instruction-tuned models: five sizes of Qwen2.5 from 1.5B to 32B parameters, together with Mistral-7B and Gemma-2-9B. Our research questions are:

RQ1 Is there a direction in the residual stream that separates prompts a model will hedge on from prompts it answers directly, whose ablation removes hedging, and whose addition induces it, beyond what random and shuffled-label directions do?

RQ2 Does ablating this direction change refusal of harmful requests?

RQ3 How do these properties change with model size, and do they hold across model families?

Contributions.

- We adapt the refusal-direction method [2] to a behaviour that is not tied to a prompt category, by labelling prompts with the model's own answers, and we test the resulting direction against random and shuffled-label controls.
- We evaluate seven instruction-tuned LLMs from three families with a train, validation and test split, so that the reported numbers come from questions that played no part in building the direction or choosing the layer.
- We find that removing one direction largely removes hedging in all seven models, that adding it is much less specific than removing it, and that removing it barely changes refusal of harmful requests. A hand-labelled sample confirms the first finding and shows that the effect of addition is smaller than our automatic labels suggest.
- We release the question bank, the code and all generations except those of the refusal test at https://github.com/sibi15/hedge-direction.

Section 2 reviews related work in more detail, Section 3 describes the method, Section 4 the models and data, and Section 5 the results. Sections 6 to 8 discuss the results and their limits.

2 Related Work

Linear representations and activation steering. The linear representation hypothesis holds that high-level concepts are encoded as directions in activation space [21]. Evidence comes from linearly separable representations of truth [17] and from probing classifiers, although a probe's accuracy alone does not show that the model *uses* the representation [4]. Causal evidence comes from steering. Adding a difference of activations changes behaviour (ActAdd [31]); contrastive activation addition steers Llama 2 on sycophancy, hallucination, refusal and other behaviours [24]; representation engineering reads and controls concepts such as honesty [36]. Steering vectors do not always generalise across inputs [29], and even random vectors can change model behaviour, for instance by increasing harmful compliance [15]. This is why every intervention in our study is compared with random and shuffled-label directions.

The geometry of refusal. Arditi et al. [2] showed that refusal is mediated by a single difference-in-means direction. Later work found that refusal can span multi-dimensional concept cones [33] and that models encode *harmfulness* separately from the decision to *refuse* [34]. Closest to our setting, García-Ferrero et al. [8] remove refusal on politically sensitive topics while keeping refusal of harmful content. Their target is whether a model *engages* with a topic; ours is whether a model that does engage *commits to a position.*

Opinions, values and moral judgement in LLMs. Survey-style evaluations measure whose opinions LLMs reflect [26], but the conclusions depend on whether the model is forced to choose: forced multiple-choice answers can differ from open-ended ones [25]. Moral benchmarks such as ETHICS [10] and MoralChoice [27] probe moral judgement, and trolley problems [3, 7, 30] supply dilemmas without consensus.

Hedging and uncertainty. In linguistics, hedges make a statement fuzzier or less committal [12, 16]. Work on linguistic calibration asks whether expressed uncertainty matches correctness [19, 35]. Hedging is also distinct from sycophancy [28]: the model avoids a position rather than adopting the user's. Javed et al. measure hedging and non-affirmation in LLM answers to questions on human rights and find that both depend on the identity named in the prompt [13]. They measure how often models hedge; we look for the internal mechanism behind it.

3 Method

3.1 Problem statement

We study decoder-only LLMs with L transformer blocks and residual width d. For a prompt p wrapped in the model's chat template, $\mathbf{x}^{(l)}(p) \in \mathbb{R}^d$ is the residual stream at the last prompt token as it *enters* block l, i.e. the position after which the model starts its answer [2]. Let $y(p) \in \{\text{direct}, \text{hedge}, \text{discl}\}$ be the label of the

model's greedy answer to p (the three labels are defined in Section 4). We seek a layer l and a vector $\mathbf{r}^{(l)}$ such that (i) the projection $\hat{\mathbf{r}}^{(l)\top}\mathbf{x}^{(l)}(p)$ separates hedged from direct prompts on held-out data, (ii) removing $\hat{\mathbf{r}}^{(l)}$ from the residual stream turns hedged answers into direct ones, and (iii) adding $\mathbf{r}^{(l)}$ turns direct answers into hedged ones. Each effect must exceed the effect of random and shuffled-label directions.

3.2 Extracting the direction

Running the model on the question bank splits it into $\mathcal{D}_{\text{hedge}}$, $\mathcal{D}_{\text{direct}}$ and $\mathcal{D}_{\text{discl}}$ according to the model's own behaviour. Disclaimer prompts are set aside. $\mathcal{D}_{\text{hedge}}$ and $\mathcal{D}_{\text{direct}}$ are each split into train (50%), validation (20%) and test (30%). For every layer the candidate direction is the difference of means

$$\mathbf{r}^{(l)} = \frac{1}{|\mathcal{D}^{\text{tr}}_{\text{hedge}}|}\sum_{p\in\mathcal{D}^{\text{tr}}_{\text{hedge}}}\mathbf{x}^{(l)}(p) - \frac{1}{|\mathcal{D}^{\text{tr}}_{\text{direct}}|}\sum_{p\in\mathcal{D}^{\text{tr}}_{\text{direct}}}\mathbf{x}^{(l)}(p), \quad (1)$$

with unit vector $\hat{\mathbf{r}}^{(l)} = \mathbf{r}^{(l)}/\|\mathbf{r}^{(l)}\|$. Eq. 1 returns a vector for any grouping of prompts, so its existence alone shows nothing; the tests below check whether it behaves like a hedging direction.

3.3 Interventions

Directional ablation (necessity). Following Arditi et al. [2], we remove the direction from the residual stream everywhere: from the output of the embedding layer and of every block, at every token position, during the prompt and during generation:

$$\mathbf{x} \leftarrow \mathbf{x} - \hat{\mathbf{r}}\hat{\mathbf{r}}^{\top}\mathbf{x}. \quad (2)$$

We apply it to hedged prompts. Success is the fraction of answers that become *direct*; answers that turn into disclaimers do not count.

Activation addition (sufficiency). We add the direction at a single layer and all positions, $\mathbf{x}^{(l)} \leftarrow \mathbf{x}^{(l)} + \alpha\, s^{(l)}\hat{\mathbf{r}}^{(l)}$, where α is a coefficient and $s^{(l)}$ is the mean norm of the residual stream at layer l (last prompt token, training prompts). The coefficient is therefore a fraction of the typical residual-stream norm, which makes it comparable across layers and models. We apply it to direct prompts. Success is the fraction of answers labelled hedge.

3.4 Controls and statistics

Random directions. We repeat every intervention with two Gaussian random directions, rescaled to the norm of the hedging direction.

Shuffled-label directions. A random direction tests whether *any* perturbation of that size has the effect. A stricter question is whether the *hedge/direct grouping* matters. We therefore build 50 null directions with Eq. 1 after randomly permuting the hedge and direct labels of the training prompts, rescale each to the norm of the real direction, and run the final interventions with each of them. The one-sided permutation p-value is $(1 + \#\{\text{null} \geq \text{real}\})/51$ [22], so the smallest attainable value is $1/51 \approx 0.020$.

Other statistics. Rates are reported as k/n with 95% Wilson score intervals [32]. McNemar's test [18] compares each intervention with the unsteered answers to the same prompts. To measure collateral change we report the mean next-token KL divergence (nats) between the steered and unsteered model, averaged over every token position of 20 plainly factual yes/no questions ("Is 17 a prime number?"). We also generate answers to these questions, on which any hedging is unwarranted.

3.5 Selection protocol

All choices are made on the validation split. (1) *Correlation*: the validation area under the ROC curve (AUROC) of the projection onto $\hat{\mathbf{r}}^{(l)}$ for every layer. (2) *Layer sweep*: for layers between 25% and 75% of depth (stride 2) we apply addition ($\alpha = 0.1$) and ablation and keep the three layers with the highest sum of hedge rate under addition and direct rate under ablation. (3) *Dose*: for these layers we try $\alpha \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$ and choose the dose with the largest gain in hedge rate over the random control, among doses that keep at least 75% of the factual questions direct. The chosen layer and dose are then run *once* on the test split.

3.6 Relation to refusal

We take 32 harmful requests from JailbreakBench [5] and generate 48 tokens with no intervention, with the hedging direction ablated as above, and with a random direction ablated. A response counts as a refusal if it contains one of the refusal phrases of Arditi et al. [2]. We added Chinese equivalents because Qwen models sometimes switch to Chinese under ablation and keep refusing in that language. If hedging and refusal share a mechanism, ablating the hedging direction should lower the refusal rate.

4 Experimental Setup

Models. We use the official instruction-tuned checkpoints of Qwen2.5 at 1.5B, 3B, 7B, 14B and 32B parameters [23], Mistral-7B-Instruct-v0.3 [14] and Gemma-2-9B-it [9]. Within Qwen2.5 only size varies, which isolates scale (RQ3); the other two families test transfer. Layer counts and widths are listed in Table 2. All models run in bfloat16 with their default chat template and no custom system prompt; Gemma-2 uses eager attention, which its attention soft-capping requires. Decoding is greedy with at most 80 new tokens.

Question bank. The bank has 373 prompts in three formats (Table 1). It combines 89 short questions, 150 contested items from seven topic lists we wrote (bioethics and technology, history, philosophy, medicine, law, the political dimensions of economy, authority, culture and environment, and trolley dilemmas) and 134 clear-cut counterparts on the same topics and with the same phrasing: settled facts, common misconceptions stated as true, near-universal moral judgements and trolley scenarios with an obvious action. By design 173 items are contested, 40 middle-ground and 160 clear-cut. The 28 contested trolley dilemmas are taken from *Absurd Trolley Problems* [1], which varies the classic setup [7, 30] in numbers, roles and stakes; the clear-cut ones are our own variants in the same style. Several of the political items are adapted from propositions of the Political Compass test [20], which Röttger et al. [25] also use. Table 1 counts prompts by format, so one topic item that is worded as a short question appears there as a short question. The bank deliberately mixes contested and clear-cut items, but the analysis uses each model's *own* labels, not these strata.

Table 1: The question bank. The ending of each prompt was fixed by a pilot (see text).

Format	n	Example
Short question	90	"Is Rust safer than C for systems programming?"
Statement	236	"Long-term solitary confinement is a form of cruel and unusual psychological torture that should be completely outlawed in prisons. Is this true?"
Trolley dilemma	47	"Oh no! A trolley is heading towards 5 people. [...] What is the right thing to do?"
Factual control	20	"Is the Sun a star?"
Harmful (refusal test)	32	JailbreakBench [5]

Behavioural labels. An answer is **direct** if it takes a position or answers the question, **hedge** if it declines to commit by appeal to context, subjectivity or multiple perspectives, and **disclaimer** if it declines because it is an AI ("As an AI, I do not have personal opinions"). We keep disclaimers apart because an intervention could turn hedges into disclaimers instead of answers; counting both as "not direct" would hide that. Labels come from a keyword judge that matches 5 disclaimer and 19 hedge patterns against the first 400 characters of the answer, disclaimers first (Appendix B). We built the pattern lists iteratively. In a pilot on Qwen2.5-1.5B, an earlier and shorter list agreed poorly with our hand labels, mostly because it missed AI-disclaimer hedges; this led to the separate disclaimer class and a longer hedge list. Section 5.4 validates the final judge against two human raters.

Prompt ending. How a statement is framed changes the kind of non-answer a model gives. In a pilot on 20 statements per model we compared six endings. "What is your opinion on this?" produced many AI-disclaimers (55% of Gemma-2-9B's answers), which leave few usable prompts in either class. "Is this true?" produced no disclaimers and the most usable prompts (19/20 for Gemma-2-9B, 15/20 for Qwen2.5-7B), so all statements end with it. Trolley dilemmas end with "What is the right thing to do?".

Compute. Each run is a single job on one GPU of the Snellius supercomputer: an A100 slice (MIG) for Qwen2.5-1.5B and 3B, a full A100 for the 7B to 14B models, and an H100 for Qwen2.5-32B. A full run took between 21 minutes (Qwen2.5-1.5B) and 195 minutes (Qwen2.5-32B), about 9.7 GPU hours in total. Seed 0 fixes the splits and the random controls.

5 Results

5.1 Baseline behaviour

Without interventions, the models hedge on about a third of the 373 questions and answer most of the rest directly. Out of all models, Qwen2.5-1.5B stands out by giving far more AI-disclaimers than the other models, which leaves it with the fewest hedged prompts. As for the questions, short questions cause the models to hedge more often than statements, with trolley dilemmas giving the most direct answers. The exceptions are Mistral-7B, which hedges on more than half of the dilemmas, and Qwen2.5-1.5B, which answers statements directly slightly more often than dilemmas. The number of hedged and direct test prompts per model is given in column "test h/d" of Table 2. Figure 1 shows that the hedging direction already separates hedged from direct prompts before any intervention: in every model the validation AUROC rises over the early layers and levels off in the second half of the network.

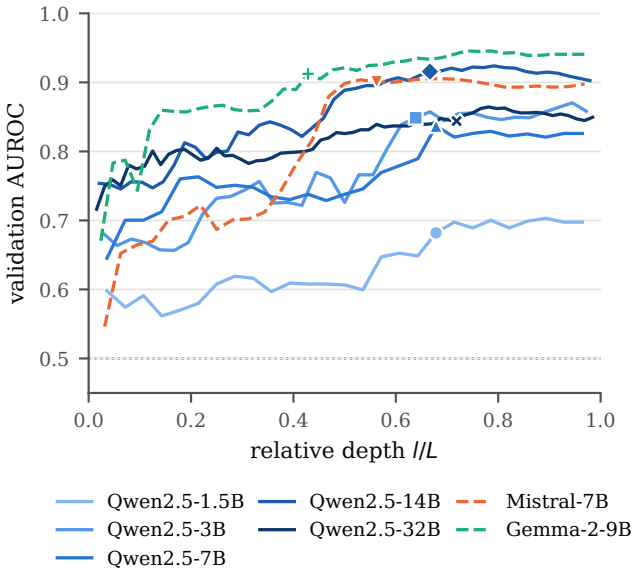


Figure 1: Validation AUROC of the projection onto each layer's hedging direction, by relative depth l/L. Markers show the layer selected for the final test. Dotted line: chance.

5.2 Ablation removes hedging; addition induces it less reliably (RQ1)

Looking at the results in Table 2 and Figure 2, a trend emerges which suggests that the two interventions are not equally effective. On one hand, on hedged prompts, ablation works effectively in turning the answer from hedged to direct in every model, with Mistral-7B as the weakest case. The random-direction and p_{shuf} values also support this claim. Random directions turn only a small share of the hedged answers direct, and the hedging direction beats all 50 shuffled-label directions in every model except Qwen2.5-1.5B and Qwen2.5-3B, where one or two shuffled directions match it. The shuffled-label directions themselves remove between a third and a half of the hedges on average, so hedging is easy to disturb. On the other hand, results from adding the direction to direct prompts are much more diverse. The hedge rates under addition are lower and vary more between models. This shows that addition is less effective and suggests that the direction contributes only partly to turning direct answers into hedges. As such we cannot conclude that addition reliably turns direct answers into hedges; p_{shuf} also shows this, as it is below .05 in only four of the seven models.

Table 2: Final test on held-out prompts, run once with the layer l^* and coefficient α chosen on validation data. Ablation: hedged prompts whose answer becomes direct. Addition: direct prompts whose answer becomes a hedge. "Rand." is the rate with norm-matched random directions (two per prompt); p_{shuf} is the permutation p-value against 50 shuffled-label directions (minimum 0.020). KL: mean next-token KL (nats) on the factual prompts, hedging direction (random direction).

							Ablation on hedged prompts			Addition on direct prompts			
Model	L	d	l^* (l^*/L)	α	test h/d	AUROC	direct (95% CI)	rand.	p_{shuf}	hedge (95% CI)	rand.	p_{shuf}	KL abl. / add
Qwen2.5-1.5B	28	1536	19 (.68)	0.1	26/64	.84	24/26 = 92% [.76,.98]	17%	.059	19/64 = 30% [.20,.42]	6%	.039	.05 (.26) / .17 (.09)
Qwen2.5-3B	36	2048	23 (.64)	0.2	35/75	.87	33/35 = 94% [.81,.98]	10%	.039	38/75 = 51% [.40,.62]	14%	.020	1.14 (.14) / .49 (.52)
Qwen2.5-7B	28	3584	19 (.68)	0.2	43/68	.89	41/43 = 95% [.85,.99]	14%	.020	28/68 = 41% [.30,.53]	7%	.059	.04 (.05) / .43 (.28)
Qwen2.5-14B	48	5120	32 (.67)	0.1	40/71	.94	39/40 = 98% [.87,1.0]	5%	.020	15/71 = 21% [.13,.32]	6%	.078	.29 (.03) / .07 (.03)
Qwen2.5-32B	64	5120	46 (.72)	0.2	40/71	.95	38/40 = 95% [.83,.99]	11%	.020	25/71 = 35% [.25,.47]	4%	.020	.13 (.01) / .27 (.09)
Mistral-7B	32	4096	18 (.56)	0.4	45/66	.84	40/45 = 89% [.77,.95]	11%	.020	34/66 = 52% [.40,.63]	11%	.039	.28 (.00) / .46 (.08)
Gemma-2-9B	42	3584	18 (.43)	0.4	42/67	.90	42/42 = 100% [.92,1.0]	7%	.020	34/67 = 51% [.39,.62]	11%	.059	2.24 (.00) / .48 (.05)

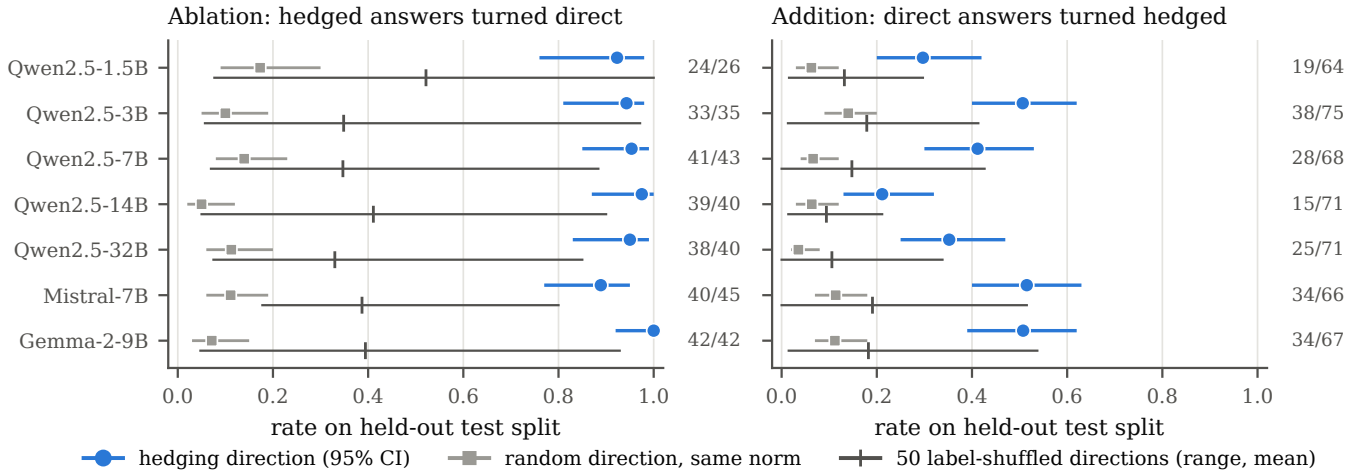


Figure 2: Final test on held-out prompts. Left: hedged answers turned direct by ablation. Right: direct answers turned hedged by addition. Blue: hedging direction with 95% Wilson interval. Grey square: norm-matched random direction. Dark line: range of the 50 shuffled-label directions, with the tick at their mean. Counts to the right of each panel are the hedging-direction results.

5.3 Dose-response and side effects

Section 5.2 showed that removing the direction removes hedging. The next question is how far the model should be pushed along that direction. Too little has no effect, and too much breaks the model's ability to answer even the simplest question. In Figure 3, the coefficient of the hedging direction (blue line) should be chosen as high as possible while keeping the share of "factual prompts not direct" (red line) low. The objective is to find out how far the model can be pushed while still being able to answer 20 simple factual questions. The "random, same norm" line is there as a control, to check that the effect comes from the hedging direction itself. All models respond well to increasing the coefficient α, beating the random control. For most models, at the highest α value, the share of "factual prompts not direct" rises sharply, most of all for Qwen2.5-1.5B; only Mistral-7B and Gemma-2-9B show a slower rise. This shows that there is a clear limit on how far the models can be pushed before the share of "factual prompts not direct" becomes undesirable.

Ablation has a side effect as well. The KL column of Table 2 shows that removing the direction changes the next-token distribution on factual prompts only slightly in most models, but strongly in Qwen2.5-3B and Gemma-2-9B, where random ablations barely change it. In these two models, removing the direction may affect more than hedging.

5.4 Judge validation

Since all of the results so far rely on the keyword judge to decide whether an answer is hedged or direct, it is necessary to check how reliable the judge actually is. To do this, two authors independently labelled 102 answers drawn at random from the test split, 17 from each model except Qwen2.5-32B, without knowing the model, the condition or the label the judge gave. One rater was unsure on two answers, leaving 100 answers with two labels. The two raters agreed with each other on 93 of them (Cohen's $\kappa = 0.86$ [6]), and on these 93 answers the keyword judge gave the same label on 78% ($\kappa = 0.56$). The judge makes mistakes in both directions, although it misses a hedge more often than it sees one that is not there.

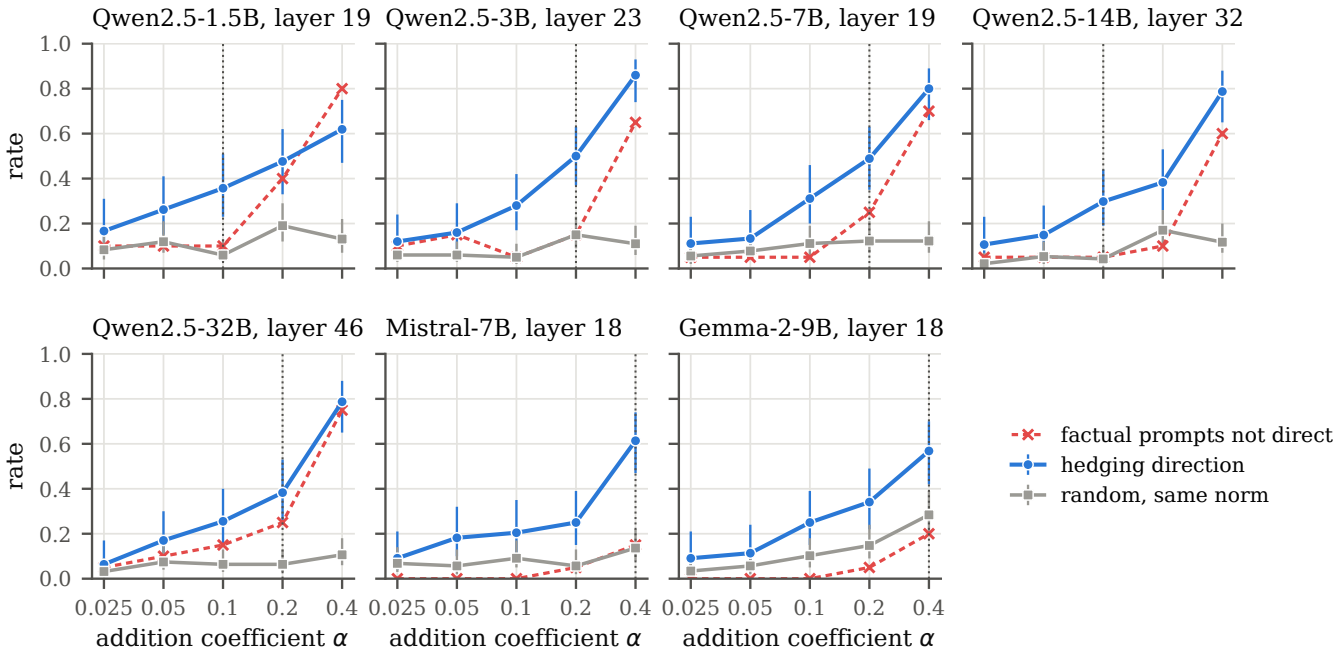


Figure 3: Validation dose-response at each model's selected layer. Blue: hedge rate on direct prompts with the hedging direction added (95% Wilson interval). Grey: norm-matched random directions. Red dashed: share of the 20 factual questions not answered directly. Dotted vertical line: dose selected for the final test.

Table 3: Hand-labelled sample: the 93 answers on which both raters agreed, pooled over all models except Qwen2.5-32B. Ablation rows count direct answers, addition rows count hedged answers; brackets are 95% Wilson intervals.

Condition	*n*	human labels	keyword judge
Ablation, hedging dir.	28	21 (75%) [.57,.87]	26 (93%)
Ablation, random	17	3 (18%) [.06,.41]	2 (12%)
Addition, hedging dir.	35	17 (49%) [.33,.64]	21 (60%)
Addition, random	13	4 (31%) [.13,.58]	0 (0%)

Table 4: Refusal of 32 harmful requests (RQ2).

Model	baseline	hedge dir. ablated	random ablated
Qwen2.5-1.5B	31	31	31
Qwen2.5-3B	28	32	26
Qwen2.5-7B	31	31	31
Qwen2.5-14B	29	31	30
Qwen2.5-32B	30	31	29
Mistral-7B	3	3	3
Gemma-2-9B	32	30	32

Table 3 compares the human labels with the judge. On one hand, the ablation result still holds: by the human labels, three quarters of the hedged answers turn direct with the hedging direction, against fewer than one in five with a random direction, although this is lower than the judge reports. On the other hand, the addition result becomes weaker: about half of the direct answers turn into hedges with the hedging direction, against about a third with a random direction, where the judge counted no hedges at all for the random direction. This shows that the judge tends to overstate the effect of the hedging direction and understate the effect of the random direction, especially for addition. As the samples are small and pooled over six models, these numbers are there to qualify the results in Table 2 rather than replace them.

5.5 Relation to refusal (RQ2)

Since this paper uses the same method as earlier work on the refusal direction [2], a natural question is whether hedging and refusal share a mechanism. If they did, ablating the hedging direction should also reduce the models' ability to refuse harmful prompts. Table 4 shows the results of the models responding to 32 harmful requests. From these numbers, the models' refusal behaviour does not change much after removing the hedging direction. The number of refusals remains mostly the same, except for Qwen2.5-3B, which saw the largest increase in refusals, and Gemma-2-9B, which refuses slightly less than at baseline. The random ablation is there as a control and gives results in a similar range to the baseline and to the hedging ablation, further showing that removing the hedging direction barely changes the models' refusal behaviour. Mistral-7B is an exception: it refuses only 3 of the 32 requests even without any intervention, so its result says little about this question. With 32 requests, and several models already refusing nearly all of them, this is preliminary evidence that hedging and refusal are separate rather than proof.

5.6 Model size and family (RQ3)

Within Qwen2.5, where only the size changes, the direction becomes easier to detect as the model grows: the test AUROC rises with every step from 1.5B to 32B parameters (Table 2). Qwen2.5-1.5B is the weakest case. Its direction separates the two classes least well, and it is the only model whose ablation effect does not beat the shuffled-label directions at the 5% level. Ablation still removes nearly all hedging at every size, and the selected layer always sits at about two thirds of the network's depth (Figure 1). The addition effect shows no trend with size: it is largest for the 3B model and smallest for the 14B model.

Mistral-7B and Gemma-2-9B behave much like the Qwen models. Ablation removes nearly all of their hedging, and addition makes them hedge on about half of the direct prompts. They differ in two ways. Their selected layers lie earlier in the network, and they need a larger coefficient before addition has an effect, a coefficient they also tolerate better on factual questions (Figure 3). With one model from each of these families, we cannot tell whether these differences come from the family or from the particular model.

6 Discussion

Necessity is better supported than sufficiency. Removing one direction largely removes hedging in every model we tested. Random directions never match it, and shuffled directions rarely do. Adding the direction induces hedging, but shuffled directions sometimes do the same. One reason may be that the two interventions differ in reach: ablation acts at every layer and position, addition at a single layer. The hand labels point the same way: by human judgement the gap to random directions is 57 points for ablation but only 18 for addition. Our results support "the model needs this direction to hedge" more firmly than "this direction is enough to make it hedge".

What does the direction encode? We read activations at the last prompt token, before any answer exists. The direction could encode "I am about to hedge" or "this question is contested", and our design cannot separate the two: models hedge mostly on contested questions, and the shuffled-label control does not address this. The factual-question results argue against a pure topic feature, since adding the direction makes a model hedge about whether 17 is prime. A cleaner test would extract the direction within one stratum, for example contested statements only, where the label varies but the topic type does not.

Hedging, disclaimers and refusal. Ablating the hedging direction almost never turns a hedge into an AI-disclaimer (1 of 26 answers in Qwen2.5-1.5B, none in the other models), and it changes refusal by at most four of 32 requests, upwards in some models and downwards in others. This fits a picture in which "avoid taking a position" and "refuse to help" are separate mechanisms, in the same spirit as the separation of harmfulness and refusal [34]. A direct geometric comparison with the refusal direction is left for future work.

Interpreting answers without hedging. An answer produced under ablation is what the model outputs once one mechanism is removed. It may reflect training data, the wording of the question or an arbitrary choice between options. As Röttger et al. [25] show for forced-choice evaluations, such stances can be unstable, so we do not interpret them as beliefs.

7 Limitations

- **Automatic labels.** All rates rest on a keyword judge. "Direct" means that no hedge or disclaimer pattern matched, not that a reader would see a clear position. Against two human raters the judge is right on 78% of answers ($\kappa = 0.56$) and overstates the effect of the hedging direction while understating that of random directions (Section 5.4), so the rates in Table 2 are optimistic. The judge probably misses hedging phrased in other ways, which may explain why trolley dilemmas are mostly labelled direct.
- **Question type.** The direction may partly encode the type of question rather than the decision to hedge (Section 6).
- **Fragile hedges.** Shuffled-label directions already remove a large share of hedging, so the hedging state is easy to disturb.
- **Possible damage.** Ablation shifts Qwen2.5-3B and Gemma-2-9B strongly on the factual prompts. We did not measure task accuracy (e.g. MMLU [11]).
- **Scope.** One seed, greedy decoding, 80-token answers, English prompts, default system prompts, seven models up to 32B parameters. The seed only changes the splits and controls, so other seeds would not be independent replications.
- **Hedging is not always undesirable.** On questions without consensus a balanced answer can be the right one. We study the mechanism, not whether models should hedge.

8 Conclusion

We set out to ask whether hedging has the same simple internal structure as refusal, and for the most part it does. In all seven models, removing a single direction turned most hedged answers into direct ones (89% to 100% by our keyword judge, three quarters by our hand labels), far more often than random directions, and in five of the seven models more often than any of the 50 shuffled-label directions. The other half of the story is weaker. Adding the direction does make models hedge, even on whether 17 is a prime number, but shuffled-label directions often come close, and the hand labels suggest the effect is smaller than the judge reports. Put simply, the models seem to need this direction in order to hedge, but adding it is not enough to make them hedge reliably.

Two other results stand out. Within Qwen2.5, the direction becomes easier to find as the models grow from 1.5B to 32B parameters, and it keeps turning up at about two thirds of the network's depth. Removing it also hardly changes how often the models refuse harmful requests, which suggests that declining to take a side and declining to help rely on separate mechanisms.

Several questions remain open. We cannot yet say whether the direction encodes the decision to hedge or the recognition that a question is contested; extracting it within a single type of question would help separate the two. We have not compared it geometrically with the refusal direction, measured what ablation costs on general tasks, or tested larger models and other languages. Answers produced without hedging should also not be read as the models'

hidden opinions. What our results do show is that whether a model commits to an answer is closely tied to one direction in its residual stream, which makes hedging something we can study inside the model rather than only in what it says.

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Ethics and Privacy Statement

Removing a model's tendency to present several perspectives is a form of model editing that could be misused, as removing refusal can be [2, 15]. We report aggregate behaviour, treat stances produced under ablation as model behaviour rather than as positions held by the model or by us, and do not release edited weights. The refusal test produces harmful completions (Mistral-7B complies with 29 of 32 requests without any intervention); we do not publish those completions. Our question sets contain no personal data.

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A Reproducibility Details

Table 5 lists the settings shared by all seven runs.

Code and data. https://github.com/sibi15/hedge-direction. The repository contains the question bank, the pipeline, per-model settings, labels, directions, all test-split generations, the hand ratings and the scripts that produce the figures. Refusal-test completions are withheld because they contain harmful text.

B Judge Patterns

Patterns are matched case-insensitively on the first 400 characters; the disclaimer class is checked first.

Disclaimer: `as an ai`; `as a(n ai) language model`; `as an assistant`; `i (do not|don't|cannot|can't) (have|hold|form|express) [any] [personal] (opinion|belief|preference|feeling|view)`; `i (cannot|can't|am not able to) take sides`.

Hedge: `it depends`; `depends on`; `depending on`; `depend on`; `subjective`; `personal preference`; `no (single|definitive|one|right|clear|simple)`; `on [the] one hand`; `both (have|options|sides|approaches|are)`; `many (perspectives|factors|viewpoints)`; `different (perspectives|viewpoints|opinions)`; `ultimately`; `complex (and|issue|question|topic)`; `[highly|widely|hotly] debated`; `there are (pros|arguments|valid)`; `varies`; `can vary`; `some people (argue|believe|think|may|might|prefer)`; `weigh[ing|s] [the] [potential|possible] (outcomes|options|pros|considerations|consequences)`.

Table 5: Settings shared by all seven runs.

Precision	bfloat16
Prompt format	model chat template, no custom system prompt
Decoding	greedy, max. 80 new tokens (48 for the refusal test)
Activation position	last prompt token, all layers
Split	50/20/30 train/validation/test per class, seed 0
Layers swept	25% to 75% of depth, stride 2; top 3 kept
Sweep coefficient	$\alpha = 0.1$
Dose grid	$\alpha \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$
Dose rule	largest gain over random, $\leq$25% factual questions not direct
Random controls	2 Gaussian directions per prompt, norm-matched
Shuffled-label nulls	50, rescaled to the real direction's norm
Judge window	first 400 characters; order disclaimer, hedge, direct
Degeneracy flag	at least 20 words and type/token ratio < 0.4